For the second object: $x_2 = v(2t) = 2x$

8 Ans: D

In 3s,
For car 1, $s = ut + \frac{1}{2} a t^2 = \frac{1}{2} a 3^2$
For car 2, $s' = ut + \frac{1}{2} a't^2 = \frac{1}{2} a'3^2$

If $a' > a$
$s' - s = 12 = \frac{1}{2} (9) (a' - a)$
$a' - a = 2.6$

In 6s, $s' - s = \frac{1}{2} (36)(a' - a) = 18(2.6) = 48$ m